

Lab 3 - Routing Protocol

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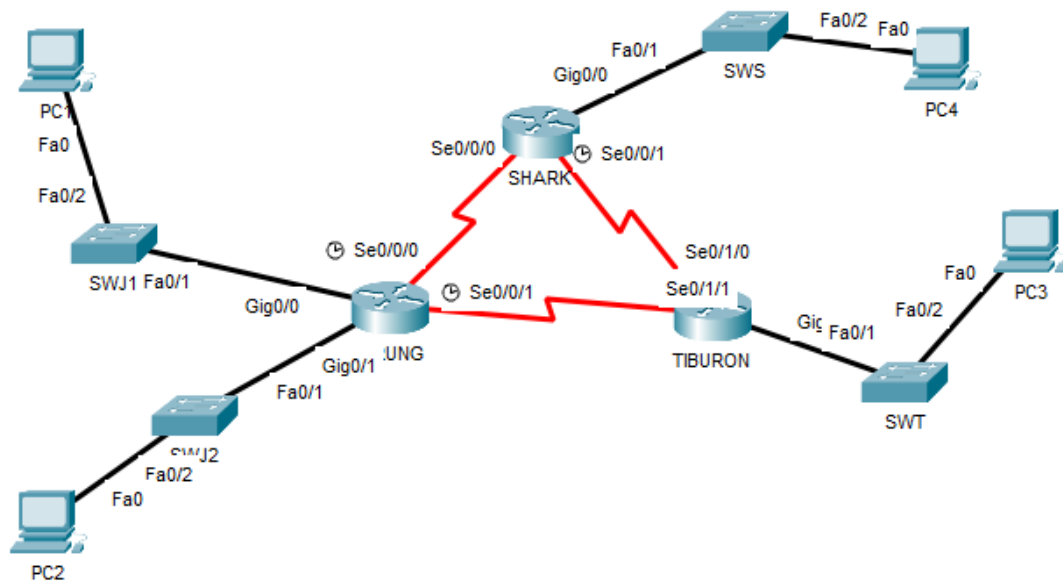
A23CS0189

SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	
2		Se0/0/1	172.18.110.197	255.255.255.252	
3		G0/0			
4		G0/1			
5	TIBURON	Se0/1/1			
6		Se0/1/0			
7		G0/0			
8	SHARK	Se0/0/0			
9		Se0/0/1	172.18.110.201	255.255.255.252	
10		G0/0			
11	PC1	-			
12	PC2	-			
13	PC3	-			
14	PC4	-			

LAB TASKS

Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	
2		Se0/0/1	172.18.110.197	255.255.255.252	
3		G0/0	172.18.110.129	255.255.255.224	
4		G0/1	172.18.110.161	255.255.255.224	
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	
6		Se0/1/0	172.18.110.202	255.255.255.252	
7		G0/0	172.18.110.65	255.255.255.192	
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	
9		Se0/0/1	172.18.110.201	255.255.255.252	
10		G0/0	172.18.110.1	255.255.255.192	
11	PC1	-	172.18.110.158	255.255.255.224	172.18.110.129
12	PC2	-	172.18.110.190	255.255.255.224	172.18.110.161
13	PC3	-	172.18.110.126	255.255.255.192	172.18.110.65
14	PC4	-	172.18.110.62	255.255.255.192	172.18.110.1

2. Using the IP addresses calculated, configure the devices with the appropriate information.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#interface GigabitEthernet0/0
SHARK(config-if)#ip address 172.18.110.1 255.255.255.192
SHARK(config-if)#no shutdown

SHARK(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/0
SHARK(config-if)#ip address 172.18.110.194 255.255.255.252
SHARK(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
SHARK(config-if)#exit
SHARK(config)#
```

```
TIBURON>enable
TIBURON#terminal configure
      ^
% Invalid input detected at '^' marker.

TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#ip address 172.18.110.65 255.255.255.192
TIBURON(config-if)#no shutdown

TIBURON(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/0
TIBURON(config-if)#172.18.110.202 255.255.255.252
      ^
% Invalid input detected at '^' marker.

TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#no shutdown

TIBURON(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

TIBURON(config-if)#exit
TIBURON(config)#interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed st
TIBURON(config)#interface Serial0/1/1
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
TIBURON(config-if)#exit
TIBURON(config)#
```

```
JERUNG>enable
JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#ip address 172.18.110.129 255.255.255.224
JERUNG(config-if)#no shutdown
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
JERUNG(config-if)#ip address 172.18.110.161 255.255.255.224
JERUNG(config-if)#no shutdown
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#ip address 172.18.110.193 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/1
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#no shutdown

JERUNG(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
exit
JERUNG(config)#
```

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

```
SHARK>enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
```

```
TIBURON>enable
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.64/26 is directly connected, GigabitEthernet0/0
L    172.18.110.65/32 is directly connected, GigabitEthernet0/0
C    172.18.110.196/30 is directly connected, Serial0/1/1
L    172.18.110.198/32 is directly connected, Serial0/1/1
C    172.18.110.200/30 is directly connected, Serial0/1/0
L    172.18.110.202/32 is directly connected, Serial0/1/0
```

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C    172.18.110.128/27 is directly connected, GigabitEthernet0/0
L    172.18.110.129/32 is directly connected, GigabitEthernet0/0
C    172.18.110.160/27 is directly connected, GigabitEthernet0/1
L    172.18.110.161/32 is directly connected, GigabitEthernet0/1
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.193/32 is directly connected, Serial0/0/0
C    172.18.110.196/30 is directly connected, Serial0/0/1
L    172.18.110.197/32 is directly connected, Serial0/0/1
```

- b. PT tool
- Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

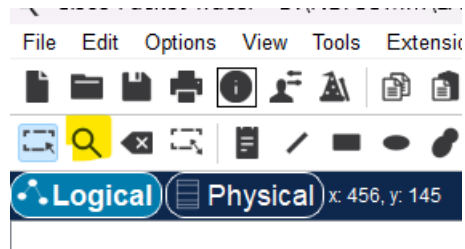


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time=11ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

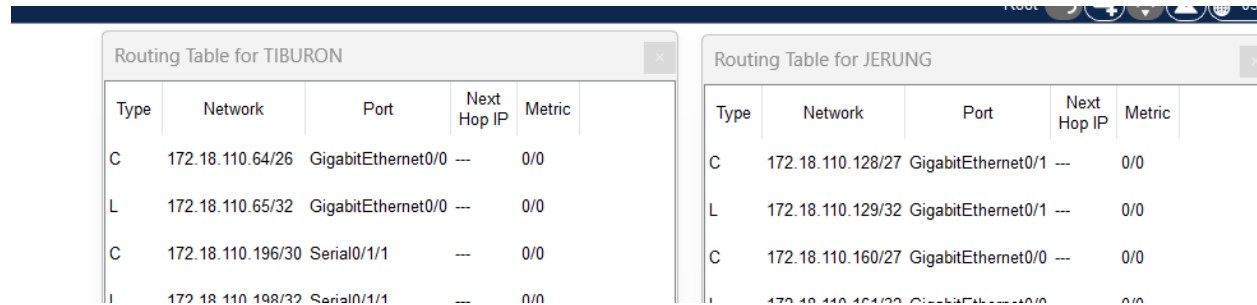
Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.
The ping to 172.18.110.190 succeeds because the destination host is correctly configured, reachable, and its interface is operable, while the ping to 172.18.110.62 fails due to a destination host unreachable, because of the 172.18.110.129 router lacks a route to the destination.
5. What needs to be done to ensure all PCs can ping each other successfully?

All routers must be operational, and the PC must be configured with the correct IP address, default gateway, and subnet mask.

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.



Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- What can you say about the addresses used in the 'network' instructions in Figure E?
The address used in the network command is the subnet id, which indicates a specific subnet in the 172.18.110.0/24 network to which the router connects and will advertise using RIP.
- Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```

TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200

```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.64/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

- What are the changes seen in TIBURON?
3 routes networks connected to JERUNG is discovered. The type R is the new route.
- What are the Networks with type R in TIBURON and JERUNG?
JERUNG: 172.18.110.64/26 , 172.18.110.200/30
TIBURON: 172.18.110.128/27 , 172.18.110.160/27 , 172.18.110.192/30
- Ping PC3 from PC1. Was it successful?

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126
Reply from 172.18.110.126: bytes=32 time=6ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 2ms
```

Successful

- d. Ping PC4 from PC1. Was it successful?

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Still not successful

- e. Explain the reasons for your answer.
[Missing routes from PC1 to PC4](#)
- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

- g. Open router SHARK's routing table and paste here.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Reply from 172.18.110.158: bytes=32 time=1ms TTL=126
Reply from 172.18.110.158: bytes=32 time=6ms TTL=126
Reply from 172.18.110.158: bytes=32 time=1ms TTL=126
Reply from 172.18.110.158: bytes=32 time=7ms TTL=126

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 3ms

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=2ms TTL=126
Reply from 172.18.110.126: bytes=32 time=4ms TTL=126
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

```

PC	Ping result
1	Success
2	Success
3	Success

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0/24](#)
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1/24

- c. After this change, can PC4 and PC1 ping PC3?
[Cannot](#)
- d. Copy and paste the routing tables for both SHARK and TIBURON here.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/16
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.
[In TIBURON the new network is already added while in SHARK the new network has not discover yet.](#)
- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
[Add the new network in RIP configuration](#)
- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```

JERUNG>enable
JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 192.168.1.0
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#exit

```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

```

C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Reply from 172.18.110.158: bytes=32 time=14ms TTL=126
Reply from 172.18.110.158: bytes=32 time=1ms TTL=126
Reply from 172.18.110.158: bytes=32 time=1ms TTL=126
Reply from 172.18.110.158: bytes=32 time=10ms TTL=126

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 6ms

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=11ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 3ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=9ms TTL=126
Reply from 172.18.110.62: bytes=32 time=10ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms

```


PC	Ping result
1	Success
2	Success
4	Success

REFLECTION

What have you learned in this task?

Through this task, I learned that the default gateway, IP address, and subnet mask must be configured accurately for communication between PCs to work. I also learned about subnetting and assigning appropriate IP addresses. In addition, I learned how to configure RIP to ensure that all routers understand each network.